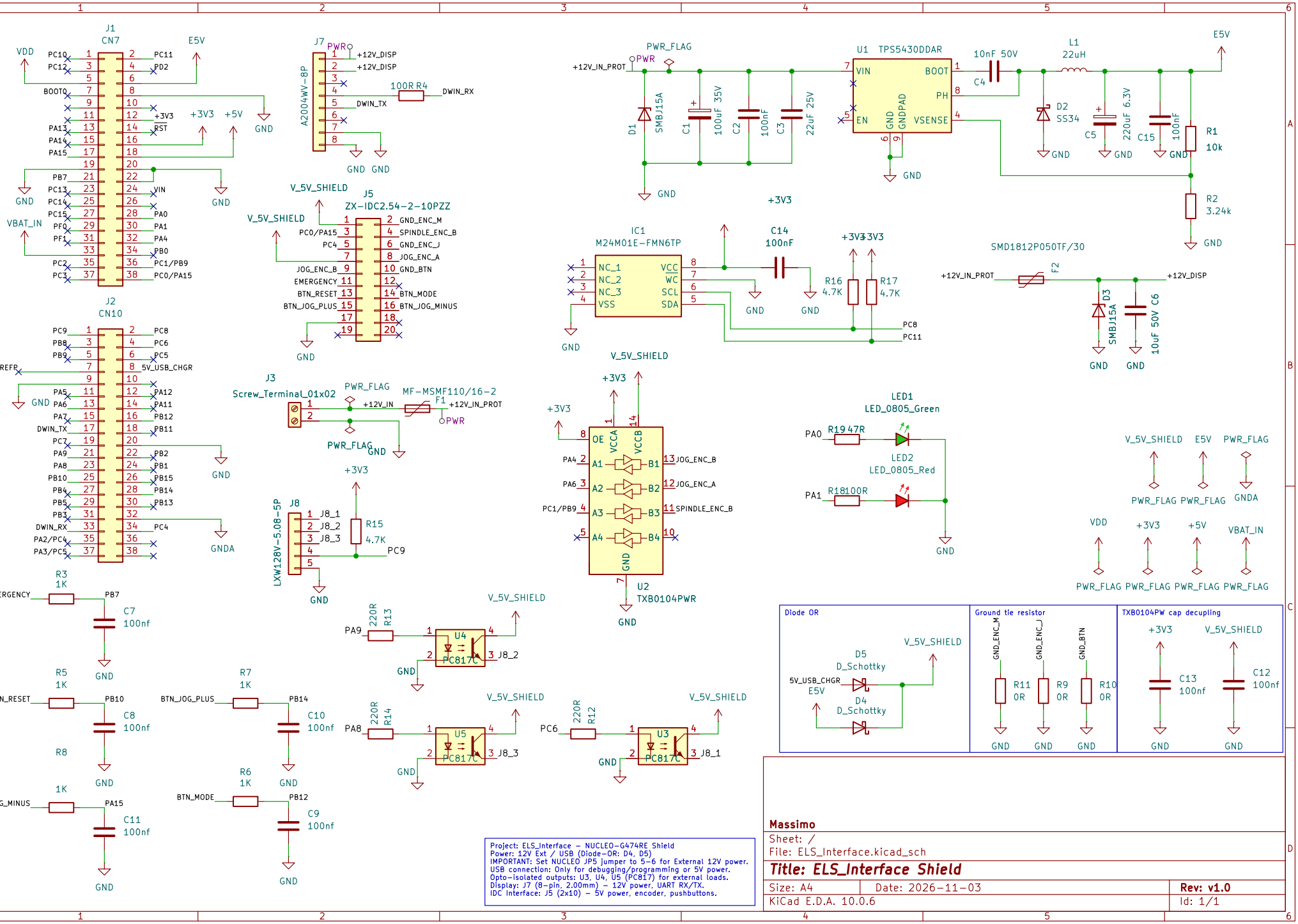

J1
CN7
J2
CN10
J7
PWR
A2004WV-8P
J5
ZX-IDC2.54-2-10PZZ
J3
Screw_Terminal_01x02
MF-MSMF110/16-2
J8
LXW128V-5.08-5P
U1
TPS5430DDAR
IC1
M24M01E-FMN6TP
U2
TXB0104PWR
U3
PC817C
U4
PC817C
U5
PC817C
LED1
LED_0805_Green
LED2
LED_0805_Red
SMD1812P050TF/30
D1
SMBJ15A
D2
SS34
D3
SMBJ15A
D4
D_Schottky
D5
D_Schottky
L1
22uH
E5V
+3V3
+5V
V_5V_SHIELD
PWR_FLAG
GND
VDD
VBAT_IN
VREFR
5V_USB_CHGR
+12V_IN_PROT
+12V_DISP
+12V_IN
GNDA
Diode OR
Ground tie resistor
TXB0104PW cap decupling
Massimo
Sheet: /
File: ELS_Interface.kicad_sch
Title: ELS_Interface Shield
Size: A4
Date: 2026-11-03
Rev: v1.0
KiCad E.D.A. 10.0.6
Id: 1/1
Project: ELS_Interface – NUCLEO-G474RE Shield
Power: 12V Ext / USB (Diode-OR: D4, D5)
IMPORTANT: Set NUCLEO JP5 jumper to 5-6 for External 12V power.
USB connection: Only for debugging/programming or 5V power.
Opto-isolated outputs: U3, U4, U5 (PC817) for external loads.
Display: J7 (8-pin, 2.00mm) – 12V power, UART RX/TX.
IDC Interface: J5 (2x10) – 5V power, encoder, pushbuttons.